# Minigraf 0507

control with Arduino Nano Every

and the Aritma XY 4131 plotter (since version 0.4.0)

from an Atari XE/XL computer (SIO) and from a PC (USB)




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© 2026 · version 0.4.1 (firmware 0.4.1)

# 1. What the interface does

The Aritma Minigraf A 0507 is a Czechoslovak pen plotter from the 1980s. It has no intelligence of its own: it only accepts a phase number for each stepper motor (X1, X2, X4 and Y1, Y2, Y4) and a pen signal. The Arduino interface does this work for the computer – it accepts readable commands (simplified HPGL), converts lines into motor steps and handles acceleration and deceleration.

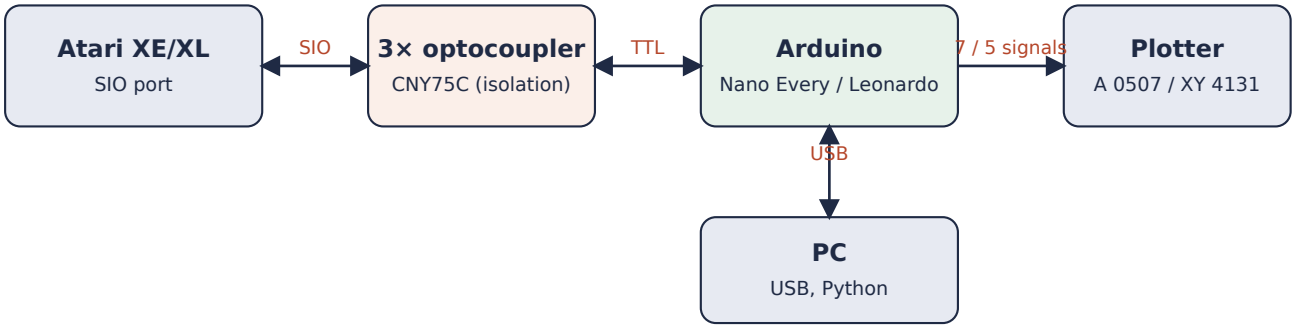



*Fig. 1 – Overall setup*

- **From the Atari** you draw with a plain LPRINT – the Arduino appears on the SIO bus as printer **P1:**.
- **From a PC** you draw with Python programs (`plot.py`, `text_plot.py`) or directly from the Arduino IDE Serial Monitor.
- The Atari is **galvanically isolated** by three optocouplers; the plotter connects to the Arduino directly.
- Transfer uses **handshaking**: the next command is sent only after the plotter has finished the previous one. Nothing gets lost.
- Since version 0.4.0 the same Arduino also drives the **Aritma XY 4131** plotter (chapter 11). Both plotters can be connected at the same time; the one selected with PL draws.

# 2. Parts list

| Part | Qty | Note |
|---|---|---|
| Arduino Nano Every (ATmega4809) | 1 | recommended; alternative Arduino Leonardo / Micro (chapter 3) |
| Optocoupler CNY75C (or CNY75A/B) | 3 | OK1 Data Out, OK2 Command, OK3 Data In |
| Resistor 1 kΩ | 3 | optocoupler LED current |
| Resistor 2.2 kΩ | 2 | pull-up on OK1 and OK2 outputs |
| Resistor 470 kΩ | 3 | base–emitter, speeds up transistor turn-off |
| SIO connector, 13 pin (or a cable from an SIO device) | 1 | pins 3, 4, 5, 7, 10 |
| Minigraf connector, 12 pin, 2.54 mm pitch | 1 | key at position 7, bottom row of contacts only |
| Optional: 7× 220 Ω, 100 µF + 100 nF | – | signal protection, Arduino supply decoupling |

## Power

- **Arduino** is powered from USB (PC or USB adapter). Do not power it from the plotter.
- **Minigraf** has its own **24 V** supply (about 30 W, supply rated at least 1.5 A). Supply minus is also the logic ground – pins 1 and 11 of the data connector.
- **The Atari side** (optocoupler LEDs) is powered from SIO pin 10 (+5 V).

**Important:** Atari ground (GND_A) and Arduino/plotter ground (GND_D) must **never be connected**. That is what the optocouplers are for.

# 3. Supported Arduino boards

The firmware `minigraf_sio.ino` detects the board from the processor. Both boards have a separate hardware UART (`Serial1` on D0/D1) for SIO, so USB stays free for the PC and debugging. All six motor phases are always on one processor port and change at the same instant.

| Signal | Nano Every | Leonardo / Micro | Minigraf pin | Function |
|---|---|---|---|---|
| SIO Data Out (via OK1) | `D0 (RX1)` | `D0 (RX1)` | – | data from Atari |
| SIO Data In (via OK3) | `D1 (TX1)` | `D1 (TX1)` | – | replies to Atari |
| SIO Command (via OK2) | `D2` | `D2` | – | start of SIO command |
| X1 | `A0` | `A0` | `10` | X phase, bit 0 |
| X2 | `A1` | `A1` | `9` | X phase, bit 1 |
| X4 | `A2` | `A2` | `8` | X phase, bit 2 |
| Y1 | `A3` | `A3` | `4` | Y phase, bit 0 |
| Y2 | `A6` | `A4` | `3` | Y phase, bit 1 |
| Y4 | `A7` | `A5` | `2` | Y phase, bit 2 |
| P (pen) | `D10` | `D10` | `5` | H = pen down |
| GND | `GND` | `GND` | `1, 11` | common ground with plotter |

| | Nano Every (recommended) | Leonardo / Micro |
|---|---|---|
| Processor | ATmega4809, 48 kB flash, 6 kB RAM | ATmega32U4, 32 kB flash, 2.5 kB RAM |
| Board package in Arduino IDE | Arduino megaAVR Boards (clones: MegaCoreX, Pinout "Nano Every") | Arduino AVR Boards (built in) |
| Motor phase port | PORTD (A0–A3, A6, A7) | PORTF (A0–A5) |
| Note | A6/A7 are full digital pins on the Nano Every | Micro uses the same pin labels |

## Minigraf connector

The cable connects to the plotter with the same type of edge connector used by ZX Spectrum bus peripherals, with **2.54 mm** pitch. It has only **12 positions**, and **position 7 is a key**. The plotter PCB has contacts on the bottom side only, so **only the bottom row** of the connector is used.




*Fig. – Connector position numbering (key at position 7 shown in black)*



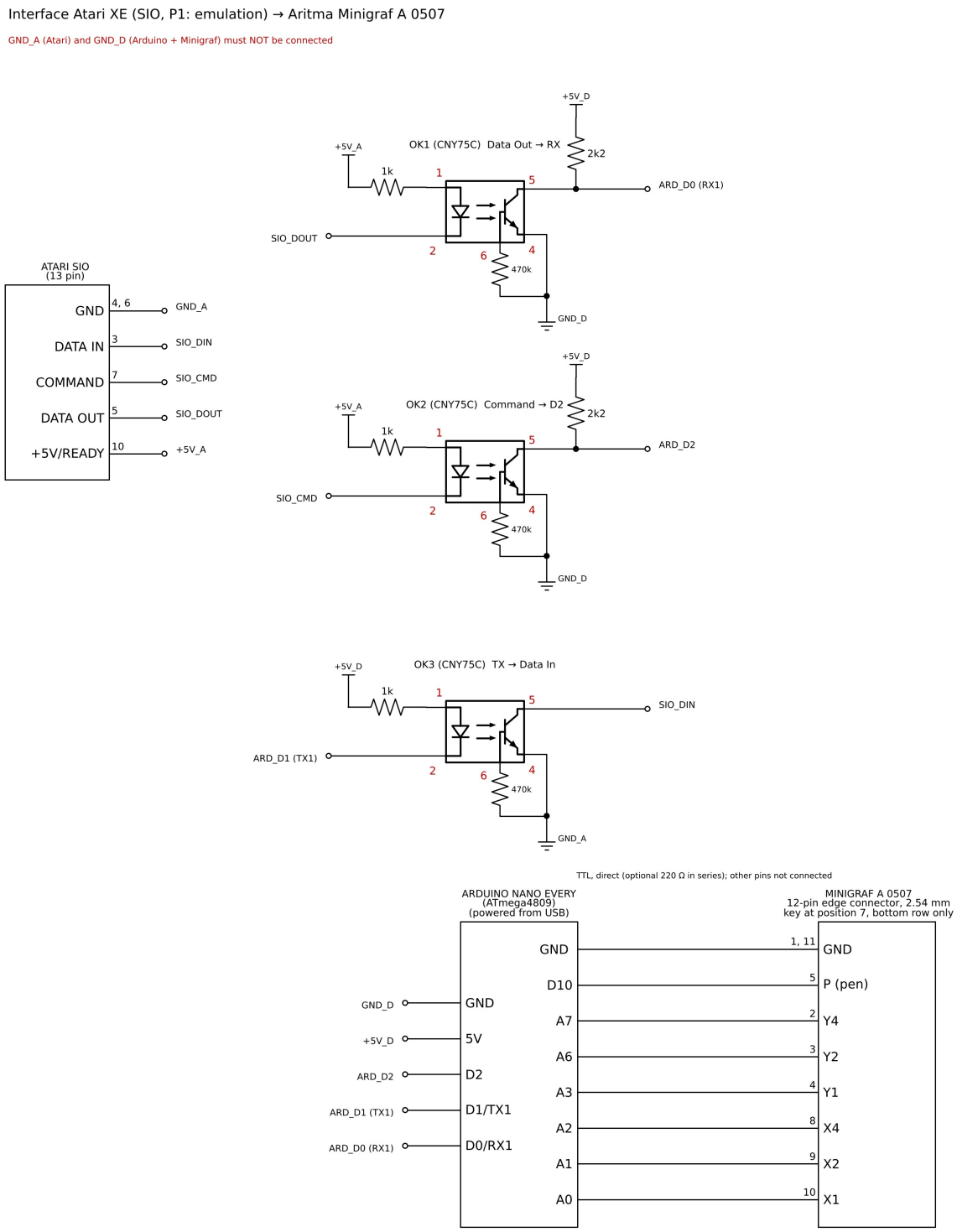

| | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| signal: | **nc** | **GND** | **X1** | **X2** | **X4** | *key* | **nc** | **P** | **Y1** | **Y2** | **Y4** | **GND** |
| Nano Every: | | GND | A0 | A1 | A2 | | | D10 | A3 | A6 | A7 | GND |

*bottom row of plotter contacts (position 7 = key, no contact)*

*Fig. – Signals at each position and the matching Arduino Nano Every pins*

> **Caution:** the key **counts as pin 7** in the numbering even though it has no contact. Contacts to the left of the key are therefore 8 to 12, not 7 to 11. Numbers moulded on some connectors do not match this numbering – check with an ohmmeter that pins 1 and 11 are plotter ground before connecting.

# 4. Schematic

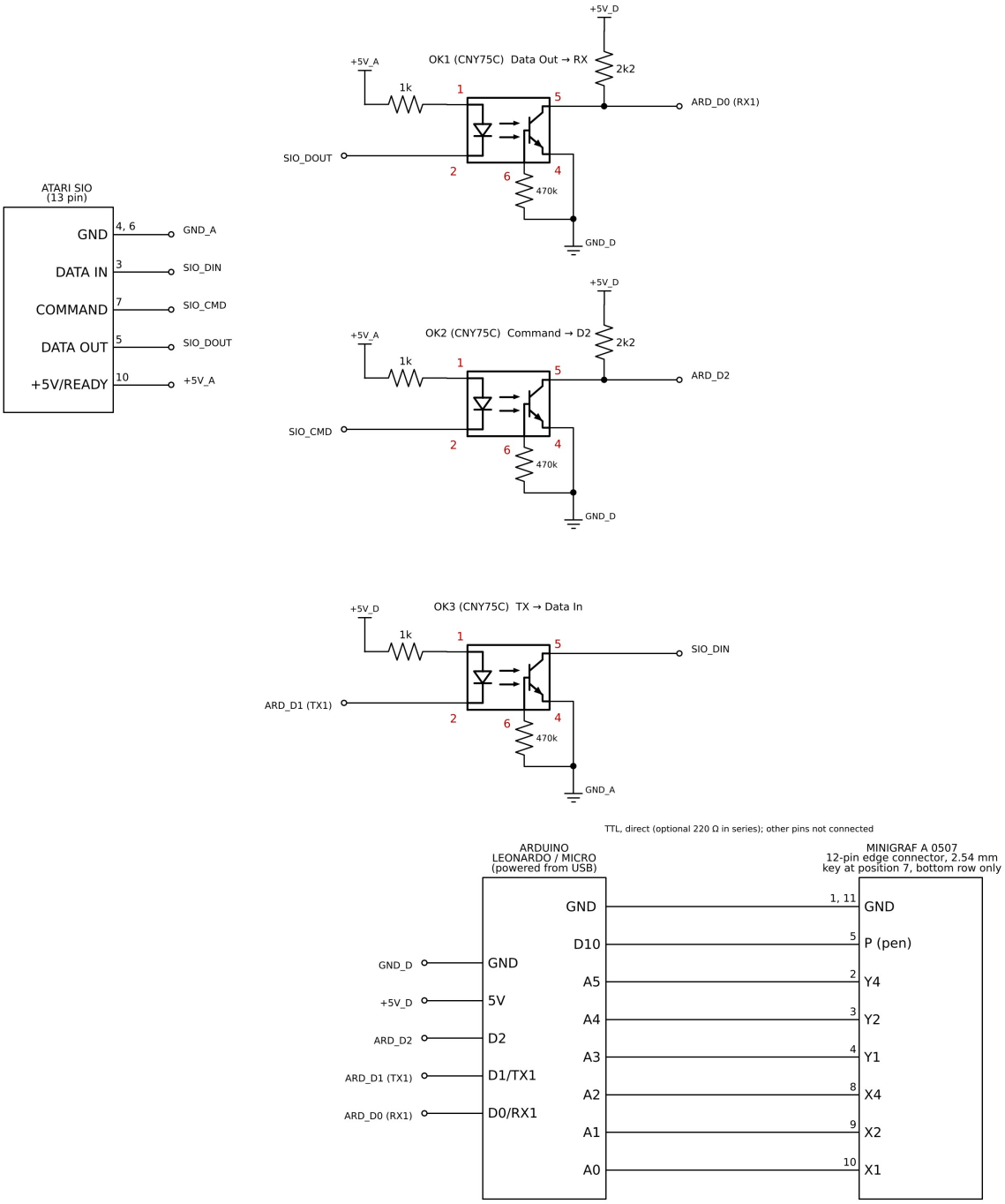



Note: +5V_A = SIO pin 10 (Atari), +5V_D = Arduino 5 V. All optocouplers are non-inverting.
If D0 is not below 0.8 V while the Atari transmits, reduce 1k at OK1/OK2 to 680 Ω.

*Fig. 2 – Wiring with Arduino Nano Every*

Interface Atari XE (SIO, P1: emulation) → Aritma Minigraf A 0507

GND_A (Atari) and GND_D (Arduino + Minigraf) must NOT be connected




Note: +5V_A = SIO pin 10 (Atari), +5V_D = Arduino 5 V. All optocouplers are non-inverting.
If D0 is not below 0.8 V while the Atari transmits, reduce 1k at OK1/OK2 to 680 Ω.

*Fig. 3 – Alternative wiring with Arduino Leonardo / Micro (only Y2 = A4 and Y4 = A5 differ)*

# 5. Software installation

## Arduino firmware

- In the Arduino IDE open the **Boards Manager**, search for **megaAVR** and install **Arduino megaAVR Boards**.
- Select **Tools → Board → Arduino megaAVR Boards → Arduino Nano Every** and the correct port.
- Open `minigraf_sio.ino` and upload it.
- The message `jtagmkII_initialize(): Cannot locate "flash" and "boot" memories` is normal and harmless on the Nano Every.
- Nano Every clone with a CH340 USB chip: use **MegaCoreX** instead, board ATmega4809 and **Pinout: Nano Every**.

## Python on the PC (Windows)

- Install Python 3 and run in the **command prompt** (not in the Python window with >>>):

```
py -m pip install pyserial
py -m pip install matplotlib      (only for --preview)
```

- Put `plot.py` and `text_plot.py` in the same folder, e.g. `C:\plotter`.
- The default port is **COM4**. Give another port before the file name or with `--port`.
- The **Serial Monitor must be closed** while plotting from Python – only one program can open the port.

# 6. Coordinates and plotting area

| Parameter | Value |
|---|---|
| Unit | 1 step = 0.125 mm (8 steps = 1 mm) |
| Absolute X position | 0 = left carriage stop (after HM), limit 1580 steps (verified, changed by LM) |
| Absolute Y position | 0 = paper loaded with its top edge at the cut-outs under the feed wheels, limit 2284 steps (verified) |
| Start position | X = 60, Y = 144 – the head moves here after HM |
| Drawing area from start | X 0 – 1520 steps (190 mm), Y 0 – 2140 steps (267.5 mm) |
| A4 margins (measured) | left and right 10 mm, top and bottom 15 mm |
| Drawing point 0,0 | set by IN; – the PC programs send HM;IN; automatically |
| Centre of area (after HM;IN;) | X = 760, Y = 1070 |

## Homing – HM (from version 0.3)

The plotter has no limit switch, but the carriage has a mechanical stop. `HM;` lifts the pen and drives the head left by the full width plus 200 extra steps. At the stop the stepper simply slips (harmless to the plotter), so the head always ends up exactly at the left, wherever it was. The firmware sets X = 0 there and moves to the start position. The Y axis (paper) has no stop: the paper position at the moment of HM is Y = 0, so A4 paper is **always loaded with its top edge at the cut-outs under the feed pinch wheels**, and HM is sent right after. The start position X = 60, Y = 144 gives A4 margins of 10 mm at the sides and 15 mm at the top. The Y limit 2284 leaves 15 mm at the bottom too – so the paper does not slip out of the feed wheels (with a longer area it slipped out by about 3 mm).

From then on the firmware knows the **absolute position** from the stop and checks the area limits against it – `IN;` only shifts the drawing origin, the limits stay in place. Limits are set with `LM x,y;` and stored in EEPROM, so they survive power-off.

The PC programs send `HM;` automatically before every drawing (disable with `--nohome`). From the Atari, start your program with `LPRINT "HM;IN;"`.

# 7. Command reference

Commands are the same from the Atari, the Serial Monitor and in PC files. Each command starts with two letters, numbers are separated by commas and commands by semicolons. Several commands may be on one line.

| Command | Meaning | Example |
|---|---|---|
| `IN;` | initialise: current position = 0,0, pen up, absolute coordinates | `IN;` |
| `PU x,y;` | pen up and move to x,y (several points allowed) | `PU760,1070;` |
| `PD x,y,...;` | pen down and draw through the given points | `PD800,100,800,900;` |
| `PA;` | following coordinates are absolute (default) | `PA;PU0,0;` |
| `PR dx,dy;` | following coordinates are relative to current position | `PR80,0;` |
| `CI r[,n];` | circle of radius r around current position, n segments (default 72) | `PU750,1050;CI400;` |
| `VS v;` | speed in mm/s; VS; = default (60 mm/s) | `VS40;` |
| `PT d,u;` | pen delay in ms after pen down / pen up; PT; = default (30 / 70) | `PT20,70;` |
| `HM[x[,y]];` | home to left stop (X = 0), paper = Y 0, then move to start (default 60,144) | `HM;` |
| `LM x[,y];` | area limits in steps from the stop, stored in EEPROM; LM; = print (default 1580,2284) | `LM1580,2284;` |
| `AC s,r;` | acceleration: start delay s µs, shortened by r µs per step; AC; = default (6000 / 250) | `AC5000,300;` |
| `DG0; / DG1;` | diagonal lines at full speed / slowed ×1.4; DG; = default (0) | `DG1;` |
| `PL0507; / PL4131;` | plotter selection, saved to EEPROM; PL; = print (since 0.4.0, chapter 11) | `PL4131;` |
| `?` | Serial Monitor only: firmware version, plotter and status (position from origin and absolute, limits, pen, speed, AC, DG, PT) | `?` |

## Useful moves

| What | Command |
|---|---|
| Home to left stop and start position | `HM;` |
| Head back to drawing origin | `PU0,0;` |
| Head to centre of area (after HM;IN;) | `PU760,1070;` |
| Move 10 mm right (without drawing) | `PU;PR80,0;PA;` |
| Square 50 × 50 mm | `PU200,200;PD600,200,600,600,200,600,200,200;PU;` |
| New origin at current position | `IN;` |

## Tuning acceleration (from version 0.2)

Every line starts slowly, accelerates to the set speed and slows down before its end. From version 0.3.2 the default is the verified setting `AC6000,250` (full speed after 2 mm) and diagonal lines run at full speed (`DG0`). For a gentler setting on an older or worn plotter use `AC9000,150;DG1;`. You can detect lost steps when the head does not return exactly to the origin after `PU0,0;`.

| Setting | Start | Full speed after | Note |
|---|---|---|---|
| `AC9000,150;` | 14 mm/s | 46 steps (5.8 mm) | gentlest (default up to 0.3.1) |

| Setting | Start | Full speed after | Note |
|---|---|---|---|
| `AC6000,250;` (default) | 21 mm/s | 16 steps (2 mm) | verified: print looks good |
| `AC5000,300;` | 25 mm/s | 10 steps (1.2 mm) | fast; check return to origin |

### Firmware defaults (Minigraf)

| Constant in minigraf_sio.ino | Value | Meaning |
|---|---|---|
| `STEP_US` | `2083` | speed 60 mm/s (= 125000 / speed) |
| `START_US / RAMP_US` | `6000 / 250` | start and stop of every line (changed by AC, verified) |
| `DIAG_SLOW` | `false` | diagonal lines at full speed (changed by DG, verified) |
| `FW_VERSION` | `"0.4.1"` | firmware version, printed at start-up and by ? |
| `X_MAX_DEF / Y_MAX_DEF` | `1580 / 2284` | default area limits from the stop (overridden by LM stored in EEPROM) |
| `HOME_X / HOME_Y` | `60 / 144` | start position after HM (A4: 10 mm left margin, 15 mm top margin) |
| `HOME_DIR` | `−1` | direction to the left stop (+1 = opposite) |
| `HOME_EXTRA / HOME_US` | `200 / 2500` | extra steps beyond the width / homing speed (~50 mm/s) |
| `MIN_STEP_US` | `1500` | maximum allowed speed about 83 mm/s |
| `PEN_DOWN_MS / PEN_UP_MS` | `30 / 70` | pen delay (verified with a 0.5 mm marker) |
| `PEN_IDLE_MS` | `1000` | pen down and no command for this long → firmware lifts it (no ink blot on a dropout) |
| `PEN_DOWN_LEVEL` | `HIGH` | P signal level for pen down |
| `X_DIR / Y_DIR` | `+1 / +1` | change to −1 to reverse an axis |
| `PRN_TIMEOUT_S` | `255` | how long the Atari waits for a line to finish |

## 8. Plotting from the Atari

On the SIO bus the Arduino behaves as printer **P1:**. You draw with a plain LPRINT in Atari BASIC (or by writing to device `P:` from other programs). After each line the Atari waits until the Arduino confirms the drawing is finished – no timing needed.

### Example Atari BASIC program

```
10 REM MINIGRAF 0507 - DEMO
20 LPRINT "HM;IN;"
30 REM FRAME
40 LPRINT "PU100,100;PD1400,100,1400,2000,100,2000,100,100;"
50 REM CIRCLE IN THE MIDDLE
60 LPRINT "PU750,1050;CI400;"
70 REM VERTICAL LINES
80 FOR I=0 TO 20
90 LPRINT "PU";100+I*60;",300;PD";100+I*60;",900;"
100 NEXT I
110 LPRINT "PU0,0;"
```

### Quick commands from the Atari

| Atari BASIC | Result |
|---|---|
| `LPRINT "HM;IN;"` | home to the stop, start position = origin |

| Atari BASIC | Result |
|---|---|
| `LPRINT "PU0,0;"` | head back to drawing origin |
| `LPRINT "VS40;"` | slower drawing (40 mm/s) |
| `LPRINT "VS;PT;"` | back to default speed and pen delay |
| `LPRINT "AC6000,250;DG0;"` | faster start and diagonal lines at full speed |
| `LPRINT "PU750,1050;CI200;"` | circle 50 mm in diameter in the middle |

## Atari tips

- One `LPRINT` may contain several commands; longer lines are split into several SIO records automatically.
- Atari BASIC prints numbers without spaces, so `LPRINT "PU";X;",";Y;";"` works directly.
- If a single line took longer than about 4 minutes, the Atari would report error 138 – split such a line into several `LPRINT`.
- While commissioning, the Serial Monitor shows `SIO: status` and `SIO: zapis OK` for each `LPRINT`.

# 9. Plotting from a PC

## 9.1 Serial Monitor (manual commands)

In the Arduino IDE open the Serial Monitor, set **115200 baud** and **Newline**. Each line you type is drawn and the Arduino replies `hotovo` ("done"). Board LED blinking = program running, steady = drawing.

## 9.2 Plotting an HPGL file – plot.py

```
py plot.py drawing.hpgl              (HPGL from Inkscape, port COM4)
py plot.py napis.hpgl --scale 1      (file already in plotter steps)
py plot.py COM7 drawing.hpgl         (another port)
py plot.py drawing.hpgl --dry        (only print what would be sent)
```

| Option | Meaning |
|---|---|
| `--scale 0.2` | coordinate scale; 0.2 = HPGL from Inkscape (40 units/mm), 1 = plotter steps |
| `--speed 40` | speed in mm/s for this drawing |
| `--pen 0,70` | pen delay in ms (after pen down, after pen up) |
| `--accel 6000,250` | acceleration for this drawing (same as AC command) |
| `--diag 0` | diagonal lines at full speed (1 = slowed) |
| `--nohome` | do not home (HM) first; draw from where the head is |
| `--dry` | send nothing, only print the converted commands |
| `--version` | print program version; the firmware version is printed automatically after connecting |

**Plotting from Inkscape:** File → Save As → *HP Graphics Language (*.hpgl)*, resolution 1016 dpi, leave "Center zero point" unticked.

## 9.3 Lettering and text – text_plot.py

The program has its own stroke font (upper and lower case, digits, punctuation, Czech diacritics). Three modes:

```
Large lettering across the whole width (A4 landscape):
  py text_plot.py "HELLO" --frame --star --bold 3 --send

Normal text in lines, wrapped automatically ( | = new line):
  py text_plot.py "Hello,|this is Minigraf." --size 5 --send
  py text_plot.py --file letter.txt --size 4 --portrait --send

Test page (6 / 4 / 3 / 2 mm):
  py text_plot.py --test --portrait --send
```

| Option | Meaning |
|---|---|
| `--send` | send to the plotter right away (otherwise only creates napis.hpgl) |
| `--size 5` | capital letter height in mm – normal text mode |
| `--portrait` | A4 portrait like a printer (lines along the short side) |
| `--file letter.txt` | text from a file (save as UTF-8) |
| `--frame / --star` | double frame / eight-pointed star above large lettering |
| `--bold 1–4` | bold: letters are traced several times with a small offset |
| `--margin 8` | margin inside the drawing area in mm (default 8 for lettering, 5 for text; the area already has A4 margins) |

| Option | Meaning |
|---|---|
| `--flip` | rotate the drawing by 180° if it comes out upside down |
| `--preview` | save a .png preview (needs matplotlib) |
| `--speed / --pen / --accel / --diag / --nohome / --port` | same as plot.py |
| `--version` | program version |

### Recommended letter size by pen

| Pen line width | Smallest well-readable letters |
|---|---|
| 0.5 mm | about 5 mm |
| 0.3 mm | about 3 mm |
| 0.1–0.2 mm | about 2 mm |

## 10. Example results




*Fig. 4 – Large lettering with frame and star, speed 80 mm/s, 0.5 mm marker.*
`py text_plot.py "ZDENĚK 0507" --frame --star --bold 3 --send --speed 80`

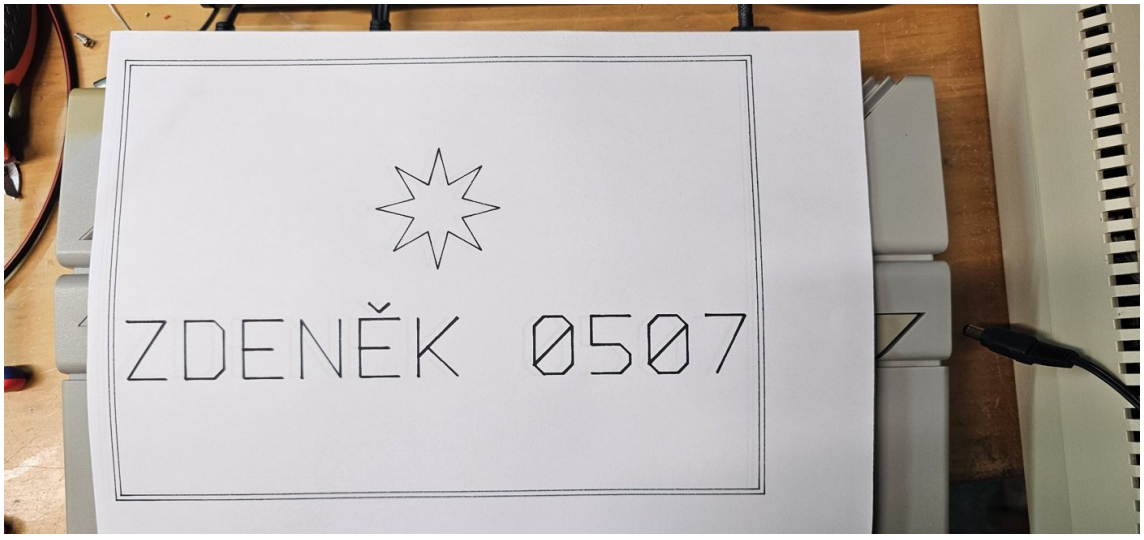
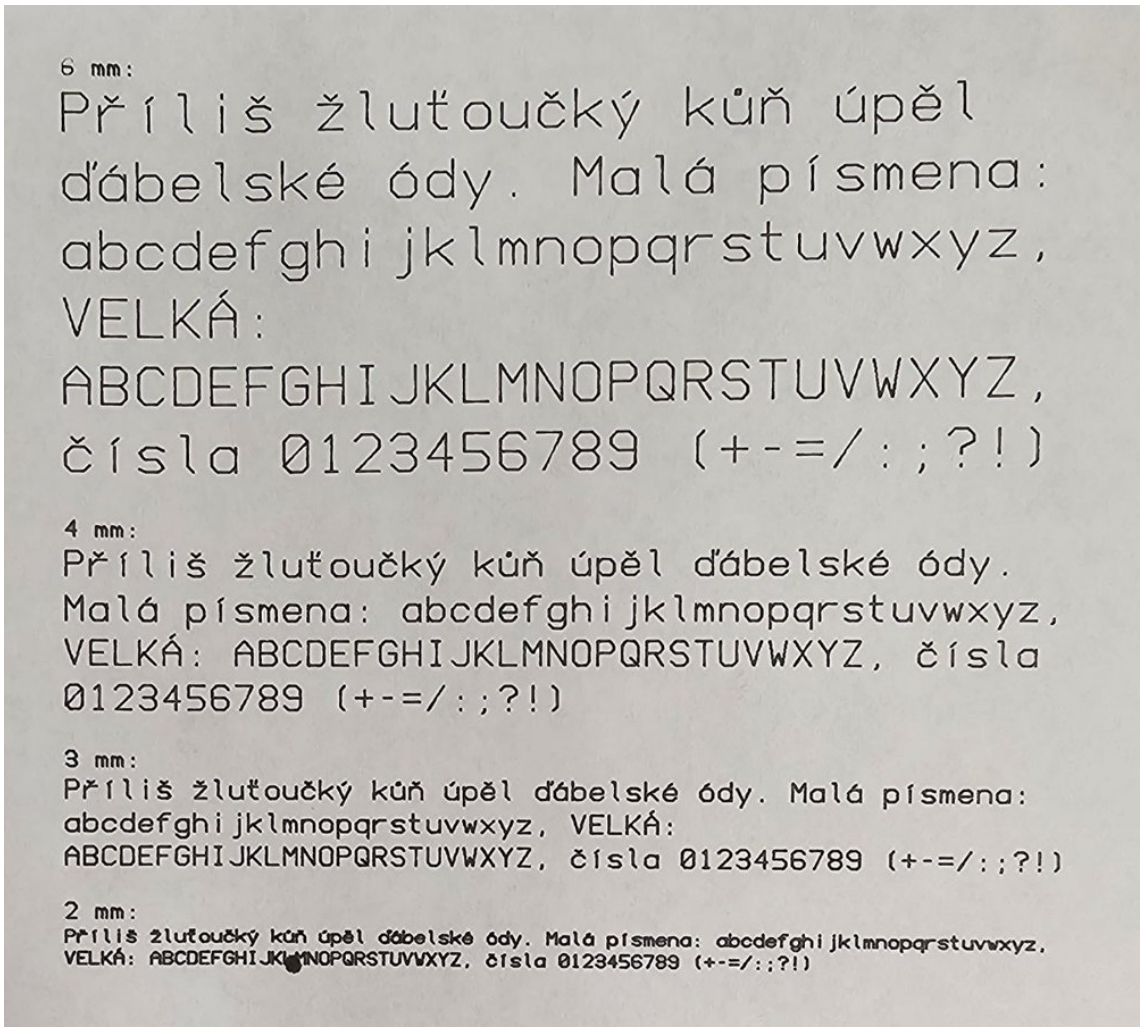



*Fig. 5 – The same drawing from the first tests, view of the plotter.*

## Fig. 6 – Test page

`py text_plot.py --test --portrait --send` – firmware and programs 0.3.4, default settings (speed 60 mm/s, acceleration AC6000,250, diagonal lines DG0, pen delay PT30,70), 0.5 mm marker, 419 s.




- 6, 4 and 3 mm: clean and easy to read, loops close, diacritics in place, lines are straight.
- 2 mm: readable, but a 0.5 mm marker is at its limit – letters look bold. A 0.3 mm marker will be clean.
- **Why the ink blot appeared** (last line near "JKLM"): at line 2058 of 2198 one command was lost on USB (`PD368,836,379,836;`), so the Arduino did not reply "hotovo" (done). Version 0.3.4 waited 20 s before querying the Arduino and resending the command – and all that time the pen stayed down on the paper, so the ink soaked into one spot. The drawing then finished correctly ("Znovu posláno řádků: 1" = 1 line resent).
- **Fix in version 0.3.5:** the firmware lifts the pen by itself when no command arrives for 1 s, and `plot.py` queries after only 3 s. A dropout now delays the drawing by just a few seconds and leaves no blot.

# 11. The Aritma XY 4131 plotter

Unlike the Minigraf, the XY 4131 (and the similar XY 4130) has its own stepper electronics. It is controlled by five 5 V logic signals: **STEP** (one pulse = 0.1 mm), **X/Y** (which axis), **+/−** (direction), **PEN** and ground. The pen carriage moves across (X axis), the paper is moved by the roller (Y axis). The plotter steps only one axis at a time, so for diagonal lines the firmware alternates the axes step by step.

## 11.1 Wiring

| Signal | Nano Every / Leonardo | WK 46248 pin | Meaning (verified) |
|---|---|---|---|
| PEN | D6 | 1 | 1 = pen down (pulled by the coil), 0 = up (spring) |
| STEP | D3 | 2 | idle 1, pulse to 0 (20 µs) = one 0.1 mm step |
| X/Y | D5 | 3 | 0 = pen carriage (X axis), 1 = paper (Y axis) |
| +/− | D4 | 4 | carriage: 1 = right, 0 = left; paper: 1 = forward, 0 = back |
| READY | D7 (not used) | 5 | open collector; always 0 on this unit, not needed by the firmware |
| GND | GND | 6 | plotter ground |
| – | – | 7 | not connected |

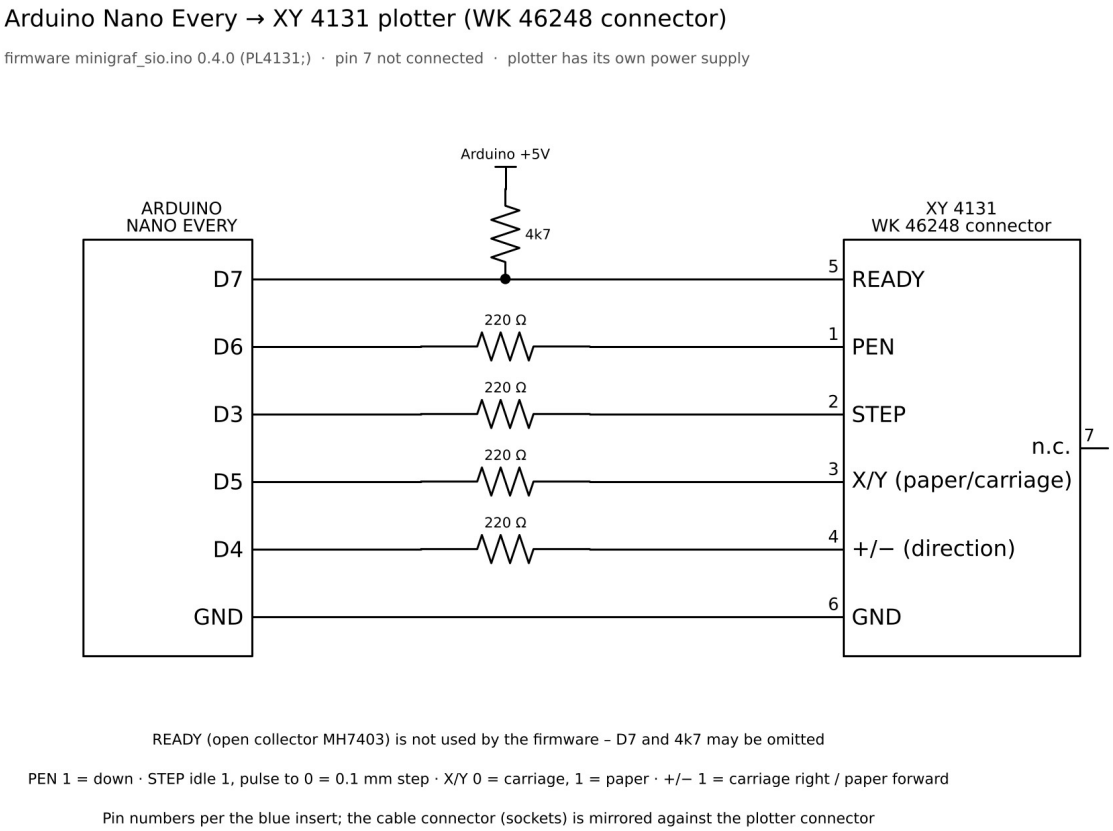



*Fig. 7 – XY 4131 wiring (220 Ω resistors recommended)*

**Check before the first drawing:** the plotter has a **pen switch** on its panel – when it is off, the pen ignores the PEN signal. The carriage end contacts must spring back freely: **a stuck left contact also blocks moving right** (the carriage then does not move even after power-cycling the plotter until the contact is freed).

## 11.2 Selecting the plotter and first start

```
PL4131;     switch to XY 4131 (saved, survives power-off)
PL0507;     back to Minigraf A 0507
PL;         print the selected plotter (also shown by ? and the start-up message)
```

After switching, the firmware loads that plotter's default speed, acceleration, pen delays and limits and treats the position as unknown – so always send `HM;` before drawing (the PC programs do it themselves). `HM;` lifts the pen, drives the carriage left over the whole travel (the end contact stops it), after a 200 ms pause sets X = 0, takes the current paper position as Y = 0 and moves to the start X = 100 (10 mm).

| Parameter | XY 4131 |
|---|---|
| Unit | 1 step = 0.1 mm (10 steps = 1 mm), measured on both axes |
| Carriage travel | about 1800 steps between the end contacts; X limit = 1780 |
| Paper | Y limit = 2600 (260 mm) from the position at HM; positive Y = paper moves back |
| Start after HM | X = 100, Y = 0 |
| Area for text_plot.py | 1600 × 2400 steps (160 × 240 mm) from the start |
| Speed | default 50 mm/s (VS; = 50), maximum 66 mm/s |
| Acceleration | AC5000,250 |
| Pen | PT30,70 – the coil pulls it down at once, the spring lifts it and it settles |

## 11.3 Drawing from a PC

`plot.py` and `text_plot.py` 0.4.0 ask the Arduino which plotter is selected after connecting and use the right scale and area automatically. Load the paper so that the pen starts about 2 cm from the edge with at least 25 cm of paper beyond it.

```
py text_plot.py "Zdeněk XY4131" --frame --star --bold 3 --send
py text_plot.py --test --send --speed 50
py plot.py picture.hpgl                (HPGL from Inkscape: scale 0.25 automatically)
py text_plot.py "HELLO" --plotter 4131 (file only, no connection)
```

## 11.4 Drawing from the Atari

Same as with the Minigraf, only the unit is 0.1 mm. Select the plotter once from the Serial Monitor (`PL4131;`) or from the Atari: `LPRINT "PL4131;HM;IN;"`.

```
10 LPRINT "PL4131;HM;IN;"
20 LPRINT "PU100,100;PD1500,100,1500,2300,100,2300,100,100;"
30 LPRINT "PU800,1200;CI500;"
40 LPRINT "PU0,0;"
```

## 11.5 XY 4131 firmware defaults

| Constant in minigraf_sio.ino | Value | Meaning |
|---|---|---|
| XY_X_DIR / XY_Y_DIR | +1 / −1 | axis directions (with XY_Y_DIR = +1 the drawing came out mirrored) |
| XY_X_MAX_DEF / XY_Y_MAX_DEF | 1780 / 2600 | area limits (changed by LM, stored separately per plotter) |
| XY_STEP_US | 2000 | speed 50 mm/s (= 100000 / speed) |
| XY_START_US / XY_RAMP_US | 5000 / 250 | acceleration and deceleration |
| XY_HOME_X / XY_HOME_Y | 100 / 0 | start after HM |
| XY_HOME_EXTRA / XY_HOME_US | 120 / 1800 | HM drives left X limit + 120 steps / speed |

| Constant in minigraf_sio.ino | Value | Meaning |
|---|---|---|
| XY_PEN_DOWN_MS / XY_PEN_UP_MS | 30 / 70 | pen delays |
| XY_PULSE_US | 20 | STEP pulse width |

### 11.6 Test program xy4131_test.ino

For bringing up a new unit the repository contains a separate program `xy4131_test.ino` (version 0.9). Lower-case commands drive single signals (`p1, a0, d1, s 1000, t 1500, h, ?`), upper-case commands draw like the main firmware. This is how the wiring, axis directions, the 0.1 mm step (1000 steps = 100 mm) and the carriage travel were verified.

## 12. Troubleshooting

| Symptom | Cause and fix |
|---|---|
| Head stops and hums | motor is losing steps – lower the speed (--speed or VS), check the 24 V supply and mechanics |
| plot.py prints "bez odpovědi, posílám znovu" (no reply, resending) | USB communication dropout (up to 0.3.2 mainly an overflow of the 64-byte Arduino buffer, fixed in 0.3.4); plot.py recovers by itself (after 3 s it queries the Arduino and resends the line) and firmware 0.3.5 lifts the pen meanwhile, so no ink blot appears. If frequent: 100 µF + 100 nF on Arduino 5 V, shorter ground to the plotter, short USB cable, ferrite on USB |
| plot.py reports "ARDUINO SE RESTARTOVALO" (restarted) | the Arduino restarted and the position is lost – run the drawing again (HM homes to the stop); usually caused by interference or a supply dip |
| could not open port / Access denied | Serial Monitor open or wrong port number |
| No module named serial | py -m pip install pyserial |
| Nothing happens from the Serial Monitor | set "Newline" and 115200 baud (bottom right) |
| Blobs at the start of lines | shorten delay after pen down: --pen 0,70 or PT0,70; |
| Start of line missing | lengthen delay after pen down: --pen 30,70 |
| Tails or smudges between letters | lengthen delay after pen up: --pen 0,100 |
| Drawing mirrored | in minigraf_sio.ino change X_DIR or Y_DIR to −1 |
| Drawing upside down | option --flip |
| Pen does the opposite (PD lifts) | change PEN_DOWN_LEVEL to LOW |
| Nothing after LPRINT, monitor silent | check OK1 and OK2; at rest the status must show Command=H |
| "chyba souctu" (checksum error) in monitor | slow edges at OK1 – reduce 1 kΩ to 680 Ω |
| Atari reports error 138 | Arduino not responding or one line takes longer than the timeout – split the line |
| Text lines are slanted | paper loaded skewed – align it with the roller |
| Marker smears even mid-line | squashed tip – replace the marker |
| HM moves right instead of left | switch the plotter off at once; set HOME_DIR = +1 in minigraf_sio.ino |
| Head does not reach the right edge | limit too small: LM; prints it, LM1580,2284; sets it |

| Symptom | Cause and fix |
|---|---|
| Paper slips out of the feed wheels | Y limit too large: reduce it, e.g. LM1580,2250; |
| Drawing is shifted on the paper in Y | paper was not loaded at the cut-outs, or moved after HM – reload it and send HM |
| Small text is drawn slowly | faster start: --accel 6000,250 or AC6000,250; |
| plot.py reports firmware "neznámá" (unknown) | Arduino has firmware older than 0.2 – upload the new one |
| Firmware does not know HM | Arduino has firmware older than 0.3 – upload the new one, or plot with --nohome |
| XY 4131: pen does not react | switch on the pen switch on the plotter; check D6 → pin 1 (about 4.4 V after PD) |
| XY 4131: carriage moves left but not right | stuck left end contact – free it, it must spring back |
| XY 4131: drawing mirrored | flip the sign of XY_Y_DIR |
| LED not blinking, monitor silent after the start-up message, Command=L | firmware before 0.4.1 waited for the Atari when the SIO interface was disconnected – upload 0.4.1 (it then only reports "SIO Command trvale L" and USB works) |
| The wrong plotter draws | PL; prints the selected one, PL0507; / PL4131; switches |

# 13. Project files

| File | Content |
|---|---|
| `firmware/minigraf_sio/minigraf_sio.ino` | firmware for Arduino Nano Every and Leonardo/Micro (Minigraf A 0507 and XY 4131) |
| `firmware/xy4131_test/xy4131_test.ino` | test program for bringing up the XY 4131 |
| `pc/plot.py` | sending HPGL files from a PC |
| `pc/text_plot.py` | lettering, text in lines and test page |
| `docs/` | manuals (CZ, EN), schematics and photos |

*Note: the firmware and PC programs print their messages in Czech. Connector pinout based on cygnus.speccy.cz (Aritma Minigraf 0507); signal meaning from the MZXR ZX Spectrum documentation and the MAME emulator. XY 4131 wiring from the XY 4130 schematic and schotek.cz, verified by measurement.*